

GALOIS THEORY (A SHORT SUMMARY)

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The following notes were taken when I studied Galois Theory in the spring & summer of 2024. The textbook used were Milne's *Field and Galois Theory* and E. Artin's lecture notes.

The notes may contain mistakes. The concepts and statements are listed, while the proofs are either not included or informally sketched.

Contents

	2
1 Basic Definitions and Results	3
2 Splitting Fields	5
3 The Fundamental Theorem of Galois Theory	7
4 Computing Galois Groups	10
5 Applications of Galois Theory	12
6 Algebraic Closure	16

1 Basic Definitions and Results

Proposition 1.1. *The characteristic of a field is either 0 or p , where p is a prime integer.*

Definition 1.2. *If a ring R contains a field of characteristic p , it is said to have characteristic p as well.*

*Given that $\text{char } R = p$, the **Frobenius endomorphism** of R is $\alpha : R \rightarrow R$ with $\alpha(a) = a^p$. It is an automorphism if R is a finite field.*

Theorem 1.3 (Gauss's Lemma). *Let $f \in \mathbb{Z}[x]$. f is irreducible in $\mathbb{Z}[x]$ if and only if f is irreducible in $\mathbb{Q}[x]$.*

Theorem 1.4 (Eisenstein's Criterion). *Let $f = a_m x^m + \dots + a_0 \in \mathbb{Z}[x]$, then f is irreducible if there exists some prime p such that:*

- (i) $p \nmid a_m$.
- (ii) $p \mid a_j$, for every $0 \leq j < m$.
- (iii) $p^2 \nmid a_0$.

Proof. Showing that if $f = gh$ then p divides one of g or h , which cannot be true. □

Definition 1.5. *A F -homomorphism between E and E' , two extended fields of F , is a homomorphism that fixes F .*

Definition 1.6. *A field extension E/F is **finite** if $[E : F] = \dim_F E$ is finite (i.e., E is a finite-dimensional F -vector space).*

*$[E : F]$ is called the **degree** of E over F .*

Proposition 1.7. *For fields $F \subset E \subset L$, we have*

$$[L : F] = [L : E] \cdot [E : F].$$

Proof. Using results from linear algebra. □

Definition 1.8. *An extension E/F is **simple** if $E = F(\alpha)$ for some $\alpha \in E$.*

Proposition 1.9. *Given $f \in F[x]$ irreducible, $F[x]/(f)$ is a field with degree $\deg f$ over F .*

Definition 1.10. *Let $f \in F[x]$ irreducible, $E = F[\alpha] = F(\alpha)$ such that $f(\alpha) = 0$ is a **stem field**. We have $[E : F] = \deg f$ as $E \cong F[x]/(f)$.*

Definition 1.11. α is **algebraic** over F if there exists a $f \in F[x]$ such that $f(\alpha) = 0$.

α is **transcendental** over F if it is not algebraic over F .

E/F is **algebraic** if for every $x \in E$, x is algebraic over F . Otherwise it is **transcendental**.

*The **minimal polynomial** $f \in F[x]$ of α algebraic is the monic polynomial that generates the annihilators of α under the action of $F[x]$.*

Corollary 1.12. *E/F is algebraic and E is finitely generated over F if E/F is finite. If E is generated by a finite set of algebraic elements over F , then it is finite and algebraic.*

In particular, α is algebraic over F if $F(\alpha)/F$ is finite.

If E/F algebraic and L/E algebraic, then L/F is algebraic.

Proof. Let $\alpha \in L$ and $f = x^m + \dots + a_0$ be its minimal polynomial over E . We know $F[a_0, \dots, a_n]/F$ is finite and $F[a_0, \dots, a_n, \alpha]/F[a_0, \dots, a_n]$ is finite. Therefore, $F[a_0, \dots, a_n, \alpha]/F$ is finite and α is algebraic over F . □

Definition 1.13. *A real number is **constructible** if its length is constructible by ruler and compass.*

Theorem 1.14. *The set of constructible numbers is a field.*

If α is constructible, then it is algebraic and $[\mathbb{Q}[\alpha] : \mathbb{Q}]$ is a power of 2.

Definition 1.15. *A polynomial f splits in F if it can be written as a product of degree 1 polynomials of $F[x]$.*

Definition 1.16. Ω is **algebraically closed** if any of the following holds:

- (i) *For every $f \in \Omega[x]$, f splits.*
- (ii) *Every $f \in \Omega[x]$ has a root.*
- (iii) *The irreducibles are of degree 1.*
- (iv) *E/Ω finite implies $E = \Omega$.*

*If Ω/F algebraic and Ω is algebraically closed, then it is an **algebraic closure** of F .*

Theorem 1.17. *If Ω/F algebraic such that every $f \in F[x]$ splits in Ω , then Ω is algebraically closed.*

Proof. Considering any stem extension over Ω and the minimal polynomial of the adjuncted element over F . The polynomial must split in Ω . □

2 Splitting Fields

Theorem 2.1. *Let Ω/F . Let $F(\alpha)$ be a simple extension, then:*

- (i) *If α is algebraic over F with minimal polynomial f , then $\{\varphi : F(\alpha) \rightarrow \Omega\}$ is in correspondence with $\{x \in \Omega, f(x) = 0\}$.*
- (ii) *If α is transcendental over F , then $\{\varphi : F(\alpha) \rightarrow \Omega\}$ is in correspondence with $\{x \in \Omega \text{ transcendental over } F\}$.*

Theorem 2.2. *For every $f \in F[x]$, there exists a splitting field F_f over F . Also, $[F_f : F] \leq n!$ if $\deg f = n$.*

Proof. Considering the degree inductively as the roots are adjuncted to F_f . □

Example 2.3. *The splitting field of $(x^p - 1)/(x - 1)$ over $\mathbb{Q}$ is $\mathbb{Q}[\zeta]$, where ζ is a primitive p -th root.*

Theorem 2.4. *Let $f \in F[x]$, $E = F[\alpha_1, \dots, \alpha_n]$ where each α_j is a root of f . Let Ω be the splitting field of f over F . The number of F -homomorphisms $E \rightarrow \Omega$ is at most $[E : F]$, and is exactly $[E : F]$ if f has distinct roots in Ω .*

Proof. Let f_1 be the minimal polynomial of α_1 in F . We then have at most $\deg f_1 = [F[\alpha_1] : F]$ homomorphisms from $F[\alpha_1]$ to Ω , where the equality holds if f_1 has distinct roots in Ω (in particular, if f has distinct roots, as $f_1 \mid f$).

Similarly, let f_2 be the minimal polynomial of α_2 over $F[\alpha_1]$, then each $\varphi_1 : F[\alpha_1] \rightarrow \Omega$ extends to at most $\deg f_2 = [F[\alpha_1, \alpha_2] : F[\alpha_1]]$ homomorphisms from $F[\alpha_1, \alpha_2]$ to Ω . The equality holds if f has distinct roots in Ω .

Repeating this process and using the multiplicity of degrees, we have at most $[E : F]$ homomorphisms from E to Ω . □

Proposition 2.5. *Every two splitting fields of the same polynomial are isomorphic.*

Proof. Consider the degrees of two fields over F and using the above theorem. □

Theorem 2.6. *Let f, g be polynomials in $F[x]$ and Ω/F an extension, then $\gcd(f, g)$ is the same as in $F[x]$ and in $\Omega[x]$.*

In particular, two irreducible polynomials in $F[x]$ do not have common divisor in any extension of F .

Proof. Using Bézout's theorem to express the gcd. □

Proposition 2.7. *A root of f is multiple if and only if it is a root of f' .*

Theorem 2.8. *The following statements are equal for a non-constant irreducible $f \in F[x]$:*

- (i) *f has a multiple root.*
- (ii) *$\gcd(f, f') \neq 1$.*
- (iii) *$\text{char } F = p \neq 0$ and f is a polynomial in x^p .*
- (iv) *All the roots of f are multiple.*

Proof. To show that (iii) implies (iv), in some extension field, consider

$$f(x) = g(x^p) = \prod (x^p - a_j)^{m_j} = \prod (x - \alpha_j)^{pm_j}$$

where $\alpha_j^p = a_j$. □

Corollary 2.9. *f has only simple roots if and only if $\gcd(f, f') = 1$.*

Definition 2.10. f is **separable** if it is non-zero and satisfies the above conditions.

Being separable does not depend on which extension f lies in.

Proposition 2.11. Every constant polynomial is separable. A non-constant irreducible polynomial is separable if $\text{char } F = 0$ or if $\text{char } F = p$ and that f is not a polynomial in x^p .

Definition 2.12. F is **perfect** if $\text{char } F = 0$ or if $\text{char } F = p$ and that every element of F is a p -th power.

Theorem 2.13. F is perfect if and only if every irreducible polynomial f is separable.

Proof. If $a \in F$ is not a p -th power, then $x^p - a$ is irreducible and not separable, as it equals $(x - \alpha)^p$ in the splitting field.

On the other hand, if every element in F is a p -th power, then every polynomial in x^p is

$$\sum a_i x^{ip} = \sum (\alpha_i)^p x^{ip} = \sum (\alpha_i x^i)^p$$

so it is not irreducible. □

Proposition 2.14. Every finite field and every algebraically closed field is perfect.

3 The Fundamental Theorem of Galois Theory

Definition 3.1. The set of F -automorphisms of some of its extension field E forms a group by composition, it is denoted as $\text{Aut}(E/F)$.

Proposition 3.2. If E/F is a splitting field of some separable $f \in F[x]$, then $\text{Aut}(E/F)$ has order $[E : F]$.

Proof. Use the earlier lemma, which says that there are $[E : F]$ endomorphisms (and hence automorphisms by finiteness) from E to itself if f has distinct roots in E . $\square$

Definition 3.3. The **fixed field** E^G is the subfield of E containing $x \in E$ such that $g(x) = x$ for every $g \in G$.

Theorem 3.4 (Artin). If G is a subgroup of $\text{Aut}(E)$, then $[E : E^G] \leq |G|$.

Proof. (This is not trivial). As a contradiction, find a linear independent set of more than $|G|$ elements in E and try to derive a contradiction. $\square$

Theorem 3.5. If G is a finite subgroup of $\text{Aut}(E)$, then $G = \text{Aut}(E/E^G)$.

Proof. We have the following inequality:

$$[E : E^G] \leq |G| \leq |\text{Aut}(E/E^G)| \leq [E : E^G]$$

where the last inequality comes from a corollary of **Theorem 2.4**. $\square$

Definition 3.6. Let E/F algebraic.

- (i) E/F is **separable** if for every $\alpha \in E$, the minimal polynomial of α in $F[x]$ is separable.
 - (ii) E/F is **normal** if for every $\alpha \in E$, the minimal polynomial of α in $F[x]$ splits in E .
 - (iii) E/F is **Galois** if it is separable and normal and finite.
- If E/F is Galois, $\text{Aut}(E/F)$ is denoted as $\text{Gal}(E/F)$.

Corollary 3.7. If F is perfect, then every algebraic extension E over F is separable.

Theorem 3.8. For an extension E/F , then followings are equivalent:

- (i) E is the splitting field for some $f \in F[x]$ separable.
- (ii) E/F finite and $F = E^{\text{Aut}(E/F)}$.
- (iii) $F = E^G$ for some finite subgroup G of $\text{Aut}(E)$.
- (iv) E/F is Galois.

Proof. (i) $\Rightarrow$ (ii): Let $F' = E^{\text{Aut}(E/F)}$. By **Proposition 3.2** we have

$$|\text{Aut}(E/F')| = [E : F'] \leq [E : F] = |\text{Aut}(E/F)|,$$

so $\text{Aut}(E/F') = \text{Aut}(E/F)$ and so $[E : F'] = [E : F]$, implying $F = F'$.

(ii) $\Rightarrow$ (iii): Let $G = \text{Aut}(E/F)$, G is finite as E/F is.

(iii) $\Rightarrow$ (iv): Let $\{\alpha_1, \dots, \alpha_n\}$ be the orbit containing α in E under the action of G . The minimal polynomial of α is then

$$\prod (x - \alpha_j)$$

which can be shown to lie in $F[x]$. Therefore we have E/F Galois.

(iv) $\Rightarrow$ (i): Let $E = F[\alpha_1, \dots, \alpha_n]$, $f = \prod f_j$ then suffices, where f_j is the minimal polynomial of α_j over F . $\square$

Corollary 3.9. Let G be a finite subgroup of $\text{Aut}(E)$ and $F = E^G$, then E/F is Galois and $[E : F] = |G|$.

Corollary 3.10. Every finite separable extension E/F is contained in some Galois extension of F .

Corollary 3.11. If E/F and $F \subset M \subset E$, then E/M is Galois.

Proof. Let $f \in F[x]$ be the polynomial such that E is the splitting field of. Consider the same $f \in M[x]$. $\square$

Definition 3.12. A Galois extension E/F is **cyclic** if $\text{Gal}(E/F)$ is cyclic.

Remark 3.13. I guess that the most takeaway from the above theorems is that if E/F is Galois, then (E/F) has the same order as $[E : F]$. Moreover, $E^{(E/F)} = F$.

Theorem 3.14 (Fundamental Theorem of Galois Theory). Let E/F Galois, $G = (E/F)$, then we have the following:

- (i) $\{H, H \text{ subgroup of } G\}$ is in bijective correspondence with subextensions $\{F \subset M \subset E\}$. The correspondence is given by sending H to E^H and reversely sending M to $\text{Gal}(E/M)$.
- (ii) $H_1 \subset H_2 \Leftrightarrow E^{H_1} \supset E^{H_2}$.
- (iii) $[H_2 : H_1] = [E^{H_1} : E^{H_2}]$ if $H_1 \subset H_2$.
- (iv) $E^{\sigma H \sigma^{-1}} = \sigma(E^H)$. $(E/\sigma M) = \sigma(E/M)\sigma^{-1}$.
- (v) H being a normal subgroup of G implies that E^H/F is normal and Galois, and that $\text{Gal}(E^H/F) \cong G/H$. Conversely, if M/F is a normal subextension, then $\text{Gal}(E/M)$ is a normal subgroup of G .

Proof. (i) It can be shown that these are indeed inverse maps to each other, as $\text{Gal}(E/E^H) = H$ and $E^{\text{Gal}(E/M)} = M$.

(ii) The forward direction is clear. The converse is because $E^{H_1} \supset E^{H_2}$ implies that $H_1 = \text{Gal}(E/E^{H_1}) \subset \text{Gal}(E/E^{H_2}) = H_2$.

(iii) $|H_2| = |H_1| [H_2 : H_1]$, and $\frac{|H_2|}{|H_1|} = \frac{[E : E^{H_2}]}{[E : E^{H_1}]} = [E^{H_1} : E^{H_2}]$.

(iv) This is clear by direct verification.

(v) By (iv) we have $\sigma E^H = E^H$ for every $\sigma \in G$. Therefore, every $\sigma \in G$ can be viewed as an element in $\text{Aut}(E^H/F)$. The kernel of the resulting map

$$\text{Gal}(E/F) \rightarrow \text{Aut}(E^H/F)$$

is clearly H . Notice that $(E^H)^{G/H} = F$, so E^H/F is Galois and $\text{Gal}(E^H/F) \cong G/H$.

Conversely, if M/F normal, then if $M = F[\alpha_1, \dots, \alpha_n]$, then the minimal polynomial of α_j has all the roots in M , so $\sigma \alpha_j \in M$ for every $\sigma \in G$, so $\sigma M = M$, and therefore $\text{Gal}(E/M)$ is a normal subgroup by (iv). $\square$

Corollary 3.15. Let $M_1, \dots, M_n$ be subextensions of E/F Galois. Let $H_i = \text{Gal}(E/M_i)$, then $\text{Gal}(E/M_1 \dots M_n) = \bigcap H_i$.

Proof. $\prod M_i$ is the smallest subfield containing each M_j , so the corresponding Galois group should be the largest among all subgroups contained in every H_j . $\square$

Corollary 3.16. Let E/F Galois, M a subextension, then the smallest normal subextension of F that contains M is E^N , where $N = \bigcap \sigma H \sigma^{-1}$, where $H = \text{Gal}(E/M)$.

Proof. We need to find the largest normal subgroup contained in H , which is N . $\square$

Proposition 3.17. Let E/F Galois, L/F an extension, E and L both contained in some field. Then we have EL/L Galois and $E/E \cap L$ Galois.

Moreover, we have a homomorphism

$$\text{Gal}(EL/L) \rightarrow \text{Gal}(E/E \cap L)$$

by restricting a map σ to E , and this is an isomorphism.

Proof. Let E splits f separable over F , we then have EL splits f over L and E splits f over $E \cap L$, so both extensions are Galois.

Notice that in the homomorphism, $E \cap L$ is the fixed field of the image of $\text{Gal}(EL/L)$. $\square$

Corollary 3.18. $[EL : F] = \frac{[E : F][L : F]}{[E \cap L : F]}.$

Theorem 3.19. Let E_1/F and E_2/F Galois, then so are E_1E_2/F and $E_1 \cap E_2$, and

$$\text{Gal}(E_1E_2/F) \cong \text{Gal}(E_1/F) \times_{\text{Gal}(E_1 \cap E_2/F)} \text{Gal}(E_2/F).$$

In particular, if $E_1 \cap E_2 = F$, then $\text{Gal}(E_1E_2/F) \cong \text{Gal}(E_1/F) \times \text{Gal}(E_2/F)$.

Proof. If E_i splits separable f_i over F , then E_1E_2 splits $\text{lcm}(f_1, f_2)$, so E_1E_2/F is Galois. If $\alpha \in E_1 \cap E_2$, then its minimal polynomial has n roots in $E_1 \cap E_2$, so $E_1 \cap E_2/F$ is Galois.

The isomorphism is proved by counting indices. $\square$

Proposition 3.20. If E/F is quadratic, then $E = F[\sqrt{\alpha}]$ for some $\alpha \in F$.

Theorem 3.21. If α is constructible then $[\mathbb{Q}[\alpha] : \mathbb{Q}] = 2^s$ for some $s \geq 0$. Now, conversely, if $\alpha \in K \subset \mathbb{R}$ such that $[K : \mathbb{Q}] = 2^s$ and $K/\mathbb{Q}$ is Galois, then α is constructible.

Proof. A p -group is solvable, so there is a sequence of intermediate fields of $K/\mathbb{Q}$ in correspondence to the sequence of groups due to the solvability of $\text{Gal}(K/\mathbb{Q})$.

Each consecutive extension of intermediate fields is quadratic, so is by adjuncting a square root, and therefore α is constructible. $\square$

Corollary 3.22. If $p = 2^k + 1$, then $\cos \frac{2\pi}{p}$ is constructible.

Proof. $\mathbb{Q}[\cos \frac{2\pi}{p}] \subset \mathbb{Q}[e^{(2\pi i)/p}]$ has a degree dividing 2^k over F and is Galois as $\mathbb{Q}[e^{(2\pi i)/p}]/\mathbb{Q}$ is. $\square$

Definition 3.23. Let $f \in F[x]$ separable, then its splitting field F_f is Galois over F . The Galois group G_f of f refers to $\text{Gal}(F_f/F)$.

As $\text{Gal}(F_f/F)$ permutes the roots of f , G_f can be viewed as a subgroup of S_n if f has degree n .

Definition 3.24. $f \in F[x]$ is **solvable in radicals** if there exists a sequence of fields:

$$F = F_0 \subset F_1 \subset \dots \subset F_m$$

such that

- (i) F_m splits f .
- (ii) $F_i = F_{i-1}[\alpha_i]$ where $\alpha_i^{m_i} \in F_{i-1}$ for some m_i .

Theorem 3.25. Let $\text{char } F = 0$, then f is solvable by radicals if and only if G_f is solvable.

4 Computing Galois Groups

Definition 4.1. Let $f = \prod (x - \alpha_j)$, then $\Delta f = \prod_{i < j} (\alpha_i - \alpha_j)$ and the **discriminant** $Df = \prod_{i < j} (\alpha_i - \alpha_j)^2$.

Proposition 4.2. For every $\sigma \in \text{Gal}(F_f/F)$, $\sigma \Delta f = (\text{sgn } \sigma) \Delta f$ and $\sigma Df = Df$.

Therefore, $Df \in F$ and that the subfield of F_f corresponding to $G_f \cap A_n$ is $F[\Delta f]$ if $\text{char } F \neq 2$.

Therefore, $G_f \subset A_n$ if and only if $\Delta f \in F$, which happens if and only if Df is a square in F .

Proof. Clearly $G_f \cap A_n$ is the subgroup that fixes Δf by the first claim. $\square$

Theorem 4.3. Let $f \in F[x]$ be separable. f is irreducible if and only if G_f acts transitively on the roots of f .

Notice that $\deg f \mid [F_f : F] = |G_f|$, so G_f is a subgroup of S_n such that $n \mid |G_f|$ and G_f acts transitively on $[n]$.

Proof. If there are two factors of f , then any element in G_f is going to permute roots inside each factor separately, and the action would not be transitive.

Now, for a separable irreducible f , simply extend an isomorphism between $F[\alpha_i]$ and $F[\alpha_j]$ to F_f , which will be an element in G_f that sends α_i to α_j . $\square$

Example 4.4. If $\deg f = 2$, then f is irreducible implies that $G_f \subset A_2$, so it is isomorphic to S_2 .

If $\deg f = 3$, then G_f is isomorphic to either A_3 or S_3 , depending on if Df is a square in F or not.

If $\deg f = 4$, let $f = \prod (x - \alpha_j)$, $\alpha = \alpha_1\alpha_2 + \alpha_3\alpha_4$, and β, γ defined analogously.

Their stabilizers are going to be order 8 subgroups of S_4 , and therefore contains V , the order 4 subgroup.

Now, $G_f \cap V$ has the fixed field $F[\alpha, \beta, \gamma]$, and therefore, $F[\alpha, \beta, \gamma]$ is Galois over F with Galois group $G_f/G_f \cap V$.

Now, let $g = (x - \alpha)(x - \beta)(x - \gamma)$. It lies in $F[x]$, and we can look at the degree of its fixed field over F to figure out the degree of F_f over F (there are not many possibilities as $G_f \subset S_4$).

Proposition 4.5. Let f be an irreducible polynomial in $\mathbb{Q}[x]$ with $\deg f = p$ a prime. If f has exactly 2 complex roots, then $G_f = S_p$.

Proof. Notice that one element in G_f is transposing the two complex roots. Since f has degree p we have $p \mid [F_f : F] = |G_f|$, so there is an order p element in G_f , which has to be a p -cycle. It can then be shown that a transposition and a p cycle generates S_p . $\square$

Lemma 4.6. A finite subgroup of the multiplicative group of a field is cyclic.

Proposition 4.7. Every extension of a finite fields is simple.

Theorem 4.8. $\mathbb{F}_q$ is the splitting field of $x^q - x$ over $\mathbb{F}_p$, where $q = p^n$ and p is a prime. Any two such fields are isomorphic.

$\mathbb{F}_q$ is Galois over $\mathbb{F}_p$ with cyclic Galois group generated by the automorphism $\sigma(a) = a^p$.

For any $m \mid n$, there exists a unique subfield of $\mathbb{F}_q$ with p^m elements.

Proof. The fixed field of $\langle \sigma \rangle$ is exactly $\mathbb{F}_p$.

The existence is clear. For the uniqueness, look at the subgroup of the Galois group. $\square$

Proposition 4.9. Let $f \in \mathbb{F}_p[x]$ with $\deg f = d$ and be irreducible. If $d \mid n$, then f is a factor of $x^{p^n} - x$, and it appears exactly once in its factorization. Moreover, $[E_f : \mathbb{F}_p] \leq d$.

Proof. The splitting field of x^{p^n} contains a subfield of degree d over $\mathbb{F}_p$ as $d \mid n$. Therefore, it contains a degree d field over $\mathbb{F}_p$ such that f splits in it. We then have $f \mid x^{p^d} - x$ and the splitting field of f therefore have a degree no more than d . $\square$

Proposition 4.10. *There exists an algebraic closure $\mathbb{F}$ for any $\mathbb{F}_p$. In $\mathbb{F}$ there exists exactly one field $\mathbb{F}_{p^n}$ for each n , and $\mathbb{F}_{p^m} \subset \mathbb{F}_{p^n}$ if and only if $m \mid n$.*

Proof. The Galois group of $\mathbb{F}_{p^n}$ over $\mathbb{F}_{p^m}$ is generated by sending a to a^{p^m} . □

Theorem 4.11. *Let $f \in F[x]$ separable. If G_f acts on roots of f with orbits size $m_1, \dots, m_r$, then $f = f_1 \dots f_r$ with $\deg f_j = m_j$. Each f_j is irreducible.*

If F has q elements, then G_f is generated by σ sending x to x^q .

The cycle type of σ is exactly $m_1 + \dots + m_r = \deg f$.

Proof. The subset of roots of f that correspond to an irreducible factor has to be stable under the action of G . □

Theorem 4.12. *Let $f \in \mathbb{Z}[x]$ monic such that $p \nmid \Delta f$. Let $\bar{f} = \prod \bar{f}_i$, where $\bar{f}_i$ is irreducible in $\mathbb{F}_p[x]$. Then there exists a $\sigma_f \in G_f$ such that σ_f has cycle decomposition $\sigma_1 \dots \sigma_r$, where σ_j is of length $\deg \bar{f}_j$.*

5 Applications of Galois Theory

Theorem 5.1. *Let $E = F[\alpha_1, \dots, \alpha_n]$ be a finite extension. α_j separable over F for $j \geq 2$. Then there exists some $\gamma \in E$ such that $E = F[\gamma]$.*

In fact, γ have the expression $\alpha_1 + c_2\alpha_2 + \dots + c_r\alpha_r$, $c_j \in F$.

Proof. Assume that F is infinite. It suffices to prove for $r = 2$. Let f and g be the minimal polynomials of α and β , consider the splitting field of fg . Let $\alpha_1, \dots, \alpha_n$ and $\beta_1, \dots, \beta_m$ be the roots of f and g .

Now, $\alpha_i + x\beta_j = \alpha + x\beta$ has exactly one solution. Let $c \in F$ be an element that is not a solution for any i and j . We then have $E = F[\alpha + c\beta]$.

Now, notice that $g(x)$ and $f(\gamma - cx)$ have coefficients in $F[\gamma]$ and that have β as the only common root (by our construction of c). Therefore, $\beta \in F[\gamma]$ and so $\alpha \in F[\gamma]$. $\square$

Proposition 5.2. *Let $E = F[\gamma]$ be an algebraic extension of F , then there exists finitely many fields M such that $F \subset M \subset E$.*

Conversely, if E/F is finite with finitely many intermediate fields, then E/F is simple.

Proof. Let M be an intermediate field and g the minimal polynomial of γ over M . Let M' over F be generated by coefficients of g , then since g is also the minimal polynomial of M' and that $M' \subset M$, we have

$$[E : M] = \deg g = [E : M']$$

so $M = M'$. Since $g \mid f$ where f is the minimal polynomial of γ over F , we have at most finitely many such M .

For the converse, use that a finite-dimensional vector space is a finite union of proper subspaces, so there exists some $\gamma \in E$ not in any subspace, and therefore $E = F[\gamma]$. $\square$

Theorem 5.3 (Fundamental Theorem of Algebra). *$\mathbb{C}$ is algebraically closed and the algebraic closure of $\mathbb{R}$. The set of algebraic numbers is the algebraic closure of $\mathbb{Q}$.*

Theorem 5.4. *Let $(\text{char } F) \nmid n$ and E the splitting field of $x^n - 1$ over F , then E/F is Galois and that $E = F[\zeta]$ for some primitive root $\zeta^n = 1$ and that there exists an injective homomorphism sending $\text{Gal}(E/F)$ to $(\mathbb{Z}/n\mathbb{Z})^\times$ by sending σ to i if $\sigma\zeta = \zeta^i$.*

If $F = \mathbb{Q}$, then this is an isomorphism.

Proof. The extension is Galois because $x^n - 1$ is separable. The roots of $x^n - 1$ form a subgroup of $E^\times$ and therefore is cyclic and hence generated by some ζ that is a primitive root.

As $\sigma \in \text{Gal}(E/F)$ sends ζ to one of the roots of $x^n - 1$, ζ is sent to some ζ^i . σ is then determined on the rest of E , so the homomorphism is injective. $\square$

Remark 5.5. *The homomorphism might not be surjective in general. For example, if $F = \mathbb{C}$ (algebraically closed), then the image could only be $\{1\}$.*

Definition 5.6. *The **cyclotomic polynomial** Φ_n is defined to be the product of $(x - \zeta)$'s over all primitive n th root ζ . We hence have*

$$x^n - 1 = \prod_{d \mid n} \Phi_d.$$

Proposition 5.7. *Let F be a field with $(\text{char } F) \nmid n$. The followings are equivalent:*

- (i) Φ_n is irreducible.
- (ii) $[F[\zeta] : F] = \varphi(n)$.
- (iii) $\text{Gal}(F[\zeta]/F) \rightarrow (\mathbb{Z}/n\mathbb{Z})^\times$ is an isomorphism.

In particular, all cyclotomic polynomials in $\mathbb{Q}$ are irreducible.

Proof. The minimal polynomial of ζ divides Φ_n . The minimal polynomial is Φ_n if and only if $\text{Gal}(F[\zeta]/F)$ acts on the roots of Φ_n transitively as it is irreducible, so if and only if the homomorphism is an isomorphism. $\square$

Theorem 5.8. *The regular n -gon is constructible if and only if $n = 2^k p_1 \dots p_s$ where each p_j is a Fermat prime.*

Theorem 5.9 (Dedekind). *Let F be a field and G be a group. Every finite set $\{\chi_j\}$ of homomorphisms $G \rightarrow F^\times$ are linearly independent over F . In other words, $\sum \alpha_j \chi_j = 0$ if and only if $\alpha_j = 0$ for every j .*

Proof. Prove by induction. Assume that there is a minimal set of linearly dependent homomorphisms, try to deduce that there exists a smaller linearly dependent set. $\square$

Definition 5.10. *Let E/F Galois. A **normal basis** is in the form $\{\sigma\alpha, \sigma \in \text{Gal}(E/F)\}$ for some $\alpha \in E$.*

Theorem 5.11. *For any E/F Galois, there exists a normal basis.*

Proof. Let G be the Galois group. The claim is equivalent to E being a free FG -module with rank 1, as this implies the map

$$\sum a_\sigma \sigma \rightarrow \sum a_\sigma \sigma \alpha$$

generate E freely for some fixed $\alpha \in E$.

We show this by showing

$$FG \oplus \dots \oplus FG \cong E \otimes_F E \cong E \oplus \dots \oplus E,$$

where each direct sum has n terms. This would imply that $FG \cong E$ as FG -modules by Krull-Schmidt theorem.

Here, $E \otimes E$ is a FG -module by $\sigma(e \otimes y) = e \otimes \sigma(y)$. It is also an E -vector space by $a(e \otimes y) = ae \otimes y$.

For the second isomorphism, notice that we have $E \otimes E \cong (e_1 \otimes E) \oplus \dots \oplus (e_n \otimes E) \cong E \oplus \dots \oplus E$.

Now, consider $\lambda_\sigma : E \otimes E \rightarrow E$ such that $x \otimes y$ is sent to $x \cdot \sigma(y)$. It can be verified that λ_σ is E -linear and that $\lambda_\sigma(\tau z) = \lambda_{\sigma\tau}(z)$ for every $z \in E \otimes E$. Then $\{\lambda_\sigma\} = \text{Hom}_E(E \otimes E, E)$ by linear independence and checking dimensions.

Now, $\phi : E \otimes E \rightarrow EG$ by sending z to $\sum \lambda_\sigma(z) \cdot \sigma^{-1}$ is an isomorphism of E -space and of EG -module. Therefore, $E \otimes E \cong EG \cong FG \oplus \dots \oplus FG$ as FG -modules. $\square$

Definition 5.12. *Let M be a G -module. $f : G \rightarrow M$ is a **crossed homomorphism** if*

$$f(\sigma\tau) = f(\sigma) + \sigma f(\tau)$$

for every σ and $\tau \in G$. In particular $f(1) = f(1 \cdot 1) = f(1) + f(1) = 0$.

Proposition 5.13. *If G is cyclic of order n , call the generator σ , then $f : G \rightarrow M$ is determined by $f(\sigma)$ and*

$$f(\sigma) + \sigma f(\sigma) + \dots + \sigma^{n-1} f(\sigma) = 0.$$

Therefore $f : G \rightarrow M$ is in correspondence with $x \in M$ such that x satisfies this relation.

Definition 5.14. *$f : G \rightarrow M$ is **principal** if $f(\sigma) = \sigma x - x$ for some fixed $x \in M$.*

$H^1(G, M) = \{\text{crossed homomorphisms}\} / \{\text{principal crossed homomorphisms}\}$, which is a quotient Abelian group.

Corollary 5.15. *A short exact sequence of G -modules*

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

gives an exact sequence:

$$0 \longrightarrow M'^G \longrightarrow M^G \longrightarrow M''^G \longrightarrow H^1(G, M') \longrightarrow H^1(G, M) \longrightarrow H^1(G, M'')$$

Theorem 5.16. Let E/F be a Galois extension and G be its Galois group, then $H^1(G, E^\times) = 0$.

Proof. Let f be a crossed homomorphism, it is suffice to show that we have a $x \in E^\times$ such that for every $\sigma \in G$, $f(\sigma) = \frac{\sigma x}{x}$.

By Dedekind's, we know that $\sum_{\tau \in G} f(\tau)\tau \neq 0$, so there exists an $\alpha \in E$ such that $\sum f(\tau)\tau(\alpha) \neq 0$; let this sum be β .

Now, we have

$$\begin{aligned} \sigma(\beta) &= \sum \sigma(f(\tau)\tau(\alpha)) \\ &= \sum \sigma(f(\tau))\sigma\tau(\alpha) \\ &= f(\sigma)^{-1}\beta \end{aligned}$$

so in other words we have $f(\sigma) = \frac{\sigma(\beta^{-1})}{\beta^{-1}}$, so f is principal. $\square$

Definition 5.17. Let E/F be Galois. The **norm** of $\alpha \in E$ is $\text{Nm } \alpha = \sum_{\sigma \in G} \sigma(\alpha)$. $\text{Nm } \alpha$ is in F and $\text{Nm} : E^\times \rightarrow F^\times$ is a homomorphism.

Theorem 5.18 (Hilbert's Theorem 90). Let E/F be a finite cyclic Galois extension, with σ generating the Galois group. For every $\alpha \in E^\times$ with $\text{Nm } \alpha = 1$, we have $\alpha = \frac{\beta}{\sigma\beta}$ for some $\beta \in E^\times$.

Proof. Verify that α satisfies the relation in **Proposition 5.13** which implies that there is a (principal) crossed homomorphism sending σ to α . $\square$

Proposition 5.19. Let F contains a primitive n -th root. Let $E = F[\alpha]$ such that $\alpha^n \in F$ and n is the least such positive integer, then E/F is Galois with cyclic Galois group of order n .

Conversely, if E is a cyclic extension of F with order n , then $E = F[\alpha]$ for some $\alpha^n \in F$.

Proof. Notice that $\{\zeta^i \alpha\}$ give n roots for $x^n - \alpha^n$ in E . The elements in $\text{Gal}(E/F)$ are determined by $\sigma(\alpha) = \zeta^i \alpha$.

For the second statement, let σ generate $\text{Gal}(E/F)$. As $1, \sigma, \dots, \sigma^{n-1}$ are distinct homomorphisms $E^\times \rightarrow E^\times$, we have that $\alpha = \sum \zeta^i \sigma^i \gamma \neq 0$ for some γ , and now $\sigma(\alpha) = \zeta^{-1} \alpha$, which proves that $\alpha^n \in F$. $\square$

Remark 5.20. We sometimes denote $E = F[a^{1/n}]$.

Lemma 5.21. Let $f \in F[x]$ separable, $F \subset F'$, then $\text{Gal}_F(f) \supset \text{Gal}_{F'}(f)$.

Proof. Let E' be the splitting field of f over F' . Let the roots be $\alpha_1, \dots, \alpha_n$, then $E = F[\alpha_1, \dots, \alpha_n]$ is the splitting field of f over F . Every element of $\text{Gal}(E'/F')$ sends α_j to α_k for some k , so $\sigma \rightarrow \sigma|_E$ is an inclusion from $\text{Gal}(E'/F')$ to $\text{Gal}(E/F)$. Therefore, the former is a subgroup of the latter. $\square$

Theorem 5.22 (Galois's Solvability Theorem). Let $\text{char } F = 0$, $f \in F[x]$ is solvable by radicals if and only if G_f is solvable.

Proof. Let $F \subset F[\zeta] = F'$ for some large primitive n -th root ζ . Let $G \subset G_f$ be the Galois group of f over F' .

For the backward direction, let $G = G_0 \supset G_1 \supset \dots \supset G_m = 0$ be a sequence of subgroups of G such that G_i/G_{i+1} is cyclic. Let $F = F[\zeta]$ and E be the splitting field of f over F' . We then have

$$F \subset F' = F_0 \subset \dots \subset F_m = E,$$

where $F_j = E^{G_j}$. F_j is then cyclic over F_{j-1} and hence is $F_{j-1}[\alpha_j]$ for some $\alpha_j^{m_j} \in F_{j-1}$. f is then solvable in radicals by this sequence.

For the forward direction, it suffices to find some $\tilde{E}/F$ such that f splits in $\tilde{E}$ and the Galois group is solvable.

Let $F \subset F_1 \subset \dots \subset F_m$ and f splits in F_m be a sequence of f being solvable in radicals. Let $n = \prod m_j$.

Let Ω/F be Galois such that $F_m \subset \Omega$ and Ω contains ζ , an n -th primitive root. Let $G = \text{Gal}(\Omega/F)$ and $\tilde{E}$ be the Galois extension of $F_m[\zeta]$ in Ω .

E is generated over F by $\zeta, \alpha_1, \dots, \alpha_m, \sigma\alpha_1, \dots$ where $\sigma \in G$. We then have

$$F \subset F[\zeta] \subset F[\zeta, \alpha_1] \subset \dots \subset \tilde{E}$$

in which each consecutive extension is by adjoining an r -th root, $r = n$ or m_j for some j , so this is a sequence of solvable extension, implying that $\text{Gal}(\tilde{E}/F)$ is Galois. $\square$

Theorem 5.23. *If $p \in R[x_1, \dots, x_n]$ is symmetric, then $p \in R[p_1, \dots, p_n]$, where $p_1 = \sum x_j$, $p_2 = \sum x_i x_j, \dots$, $p_n = x_1 \dots x_n$.*

Proof. Give an order on the monomials. Let $cx_1^{i_1} \dots x_n^{i_n}$ be the highest monomial of p . Consider $p - cp_1^{i_1-i_2} \dots p_n^{i_n}$, this has necessarily a smaller highest monomial. Repeating this process until $p = 0$. $\square$

Theorem 5.24. *$F(x_1, \dots, x_n)/F(p_1, \dots, p_n)$ is Galois with $\text{Gal}(F(x_1, \dots, x_n)/F(p_1, \dots, p_n)) = S_n$. The former is in fact the splitting field of $g = x^n - p_1 x^{n-1} + \dots + (-1)^n p_n$.*

Definition 5.25. *A **general polynomial** of degree n is in the form $f = x^n - t_1 x^{n-1} + \dots + (-1)^n t_n \in F[t_1, \dots, t_n][x]$.*

Theorem 5.26. *Regard $f \in F(t_1, \dots, t_n)[x]$, then the Galois group of f is S_n .*

Proof. Consider $F[t_1, \dots, t_n] \rightarrow F[p_1, \dots, p_n]$. This is injective by considering the highest monomials in $m(t_1, \dots, t_n)$.

This then extends to a homomorphism $F(t_1, \dots, t_n) \rightarrow F(p_1, \dots, p_n)$, and therefore, f can be regarded as a polynomial in $F(p_1, \dots, p_n)[x]$ and hence the Galois group is S_n as f is sent to g , the polynomial in **Theorem 5.24**. $\square$

Remark 5.27. *First, for every n there exists some $f \in \mathbb{Q}[x]$ such that $G_f = S_n$ (as we could use the result above). Now, every finite group is a subgroup of some S_n , so every finite group is a Galois group over some finite extension of $\mathbb{Q}$.*

The converse is still open: is every finite group is a Galois group over $\mathbb{Q}$?

6 Algebraic Closure

Theorem 6.1. *Let Ω/F be an algebraic extension. If for every $f \in F[x]$ that is non-constant, f has a root in Ω , then Ω is an algebraic closure of F .*

Proof. It suffices to show that for every $f \in F[x]$, f splits in Ω . Let f be separable and E be its splitting field. We then have $E = F[\gamma]$ by **Theorem 5.1**. Let g be the minimal polynomial of γ , then it has a root β in Ω , and therefore there is a homomorphism $E = F[\gamma] \rightarrow F[\beta] \subset \Omega$. Since f splits in E , it splits in Ω .

Now, assume that F is not perfect with $\text{char } F = p$. Let $F' = \{x \in \Omega, x^{p^m} \in F \text{ for some } m\}$.

We then show F' is perfect and that every polynomial in $F'[x]$ splits in Ω , implying that every polynomial in $F[x]$ splits in Ω . $\square$

Theorem 6.2. *For every field F there exists an algebraic closure Ω . For every Ω algebraic closure of F and E/F algebraic, there exists a F -homomorphism $\varphi : E \rightarrow \Omega$.*

If E is also an algebraic closure, then such φ is an isomorphism.

Proof. By Zorn's lemma. It should be cautioned that an isomorphism between algebraic closures is not canonical. $\square$

Definition 6.3. Ω is **separably closed** if for every $f \in \Omega[x]$ non-constant and separable, f splits in $\Omega[x]$.

Ω is a **separable closure** of F if Ω/F is algebraic and separable and separably closed.

Theorem 6.4. *For every field F there exists a separable closure Ω . For every Ω separable closure of F and E/F separable and algebraic, there exists a F -homomorphism $\varphi : E \rightarrow \Omega$.*

If E is also a separable closure, then such φ is an isomorphism.